

ROBUST SYSTEMS DESIGN

Systems Analysis & Design

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UNIVERSIDAD DISTRITAL
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Outline

- 1 Concepts Generation & Selection
- 2 Quality Guidelines in Systems Design
- 3 Systems Architectures



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Concepts Generation

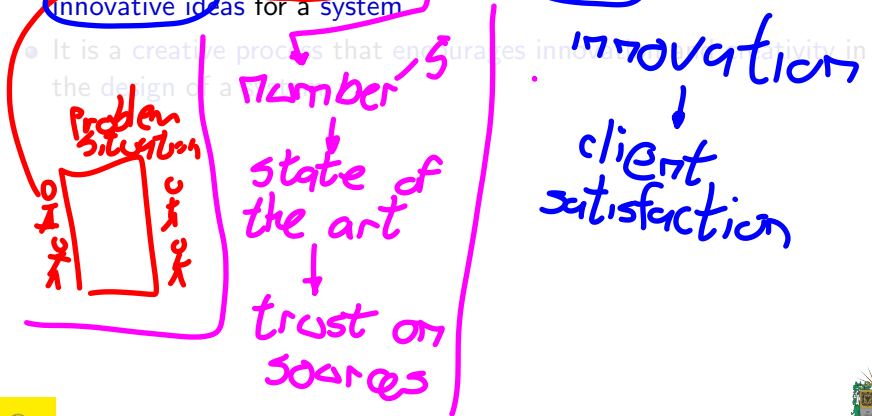
- **Concepts generation** is the process of creating ideas for a system that meet the needs of its users.
- It involves brainstorming, research, and analysis to generate innovative ideas for a system.
- It is a creative process that encourages innovation and creativity in the design of a system.

*Problem
Solving*



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team



Innovation and Creativity

- **Innovation** is the process of creating new ideas and solutions that improve the performance of a system.
- Creativity is the ability to generate original and innovative ideas that solve problems and meet the needs of users.
- They are important for ensuring that a system is robust, efficient, and effective.



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solution

resources



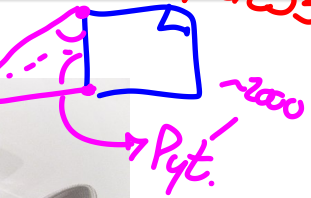
Is this Innovation & Creativity?

Wii Mote - wireless

bluetooth
(10 years before)



IR ~ 30 y



20 y before



Concepts Selection

- **Concepts selection** is the process of evaluating and choosing the best ideas for a system.
- It involves analysis, comparison, and evaluation of concepts to determine which ones are the most feasible and effective.

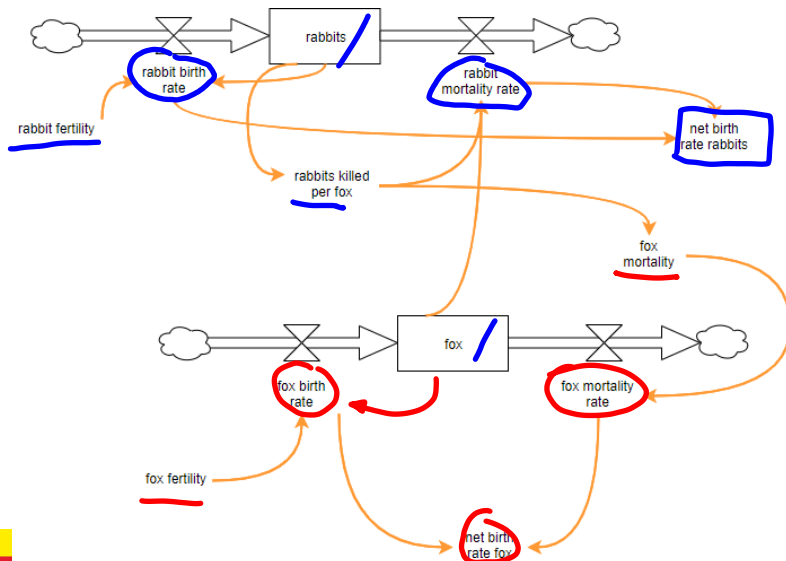
→ ideas

→ rubric

→ fair
Comparison



Stock and Flow Diagram



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Quality Guidelines

- **Quality guidelines** are principles that **guide the design** of a system to ensure that it **meets** the **needs** of its **users**.
- They include **reliability**, **scalability**, **maintainability**, and **usability** guidelines.
- They are **important** for ensuring that a **system** is **robust**, **efficient**, and **effective**.

Handwritten notes:

- RECOMMENDATION** (with arrow pointing to 'guide the design')
- test** (with arrow pointing to 'reliability')
- demand** (with arrow pointing to 'scalability')
- Fix errors** (with arrow pointing to 'maintainability')
- consequence** (with arrow pointing to 'important')
- Trade-off** (with arrow pointing to 'security vs speed')



Reliability Guidelines

- **Reliability** guidelines are *principles* that *guide* the *design* of a *system* to ensure that it is *reliable* and *dependable*.
- They include *fault-tolerance*, *redundancy*, and *error-handling* guidelines.
- They are *important* for ensuring that a *system* is *robust* and *resilient* to failures.

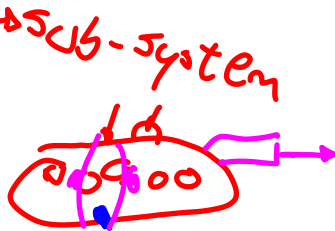
↓
happen all
the time

try-catch
logs



Scalability Guidelines

- **Scalability guidelines** are *principles* that *guide the* design of a system to *ensure* that it is *scalable* and *flexible*.
- They include *modularity*, *extensibility*, and *performance* guidelines.
- They are *important* for *ensuring* that a system can *grow* and *adapt* to changing requirements.



memory
→ executing
• response
quality
• max request



Maintainability Guidelines

- **Maintainability guidelines** are principles that guide the design of a system to ensure that it is easy to maintain and update.
- They include modularity, documentation, and versioning guidelines.
- They are important for ensuring that a system can be easily maintained and updated by its developers.

tech

connective
after client

github
lab
bitbucket

roll-back / pos



Quality Standards

- **Quality standards** are benchmarks that *define* the level of quality that a system must meet.
- They include ISO 9000, CMMI, and Six Sigma standards.

product

enterprise

process



ISO 9000

- **ISO 9000** is a quality standard that *defines* the **requirements** for a **quality management system**.
- It is *designed* to help **organizations** ensure that they **meet** the **needs** of their **customers** and **stakeholders**.
- It is *based* on a number of quality management principles, including **customer focus**, **leadership**, and **continuous improvement**.



ISO 27001

- **ISO 27001** is a quality standard that defines the requirements for an information security management system.
- It is *designed* to help organizations protect their information and ensure that it is secure and confidential.
- It is *based* on a number of information security management principles, including risk assessment, security policies, and incident response.



CMMI → J-S

- **CMMI** is a quality standard that *defines* the **requirements** for a mature software development process. → *dev. Processes*
- It is *designed* to help organizations improve their software development processes and deliver high-quality products to their customers.
- It is *based* on a number of best practices for software development, including requirements management, project planning, and process monitoring.



Six Sigma

- **Six Sigma** is a quality standard that *defines* the **requirements** for a **process** that is *capable* of **producing high-quality products**.
- It is *designed* to help **organizations** improve their **processes** and **reduce defects** in their **products** and **services**.
- It is *based* on a **number of quality management principles**, including **data-driven decision-making**, **process improvement**, and **customer focus**.



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What is a System Architecture?

- A **(system architecture)** is the **structure** of a system that defines its **components**, **interactions**, and **relationships**.
- A system architecture is the blueprint of a system that guides its development and implementation.
- A system architecture is the foundation of a system that ensures that it meets the needs of its users.

reference



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template
structure — behavior



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- A **[system architecture]** is the foundation of a system that ensures that it **meets** the **needs** of its **users**.

Solve
problem



Types of System Architectures

- There are several types of system architectures that are used in systems development.
- They include monolithic, client-server, peer-to-peer, and distributed architectures.
- Each type of architecture has its own advantages and disadvantages that depend on the specific requirements of the system.



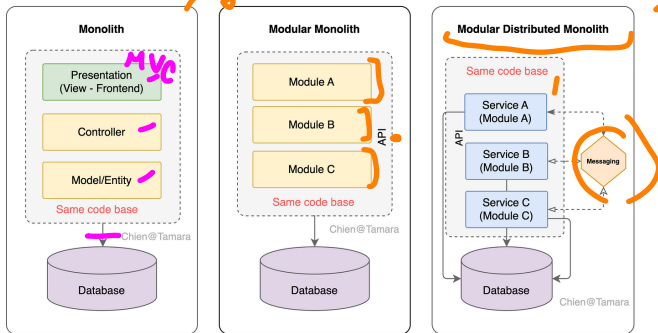
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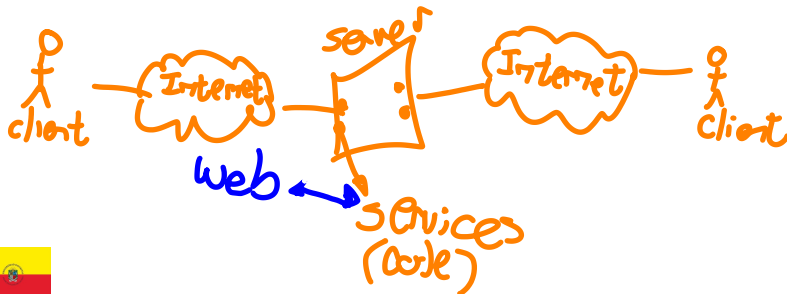
Monolithic System Architecture

- A **monolithic system architecture** is a **single-tier** architecture that consists of a **single unit** that performs all the **functions** of the system.
- It is **simple**, **easy** to develop, and **maintain**, but it is **not scalable** and **flexible**. It is **used** for **small systems** that **do not require high** performance or **reliability**.



Client-Server System Architecture

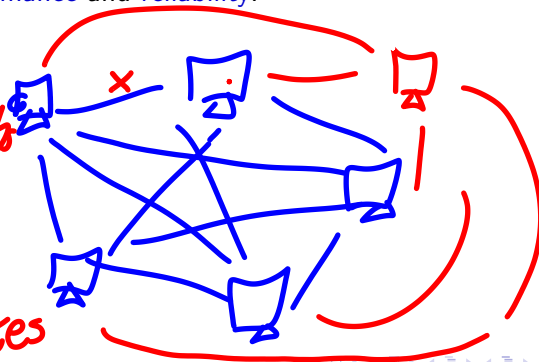
- A **client-server system architecture** is a **two-tier architecture** that consists of a **client** and a **server** that *communicate* with each other over a **network**. ① ②
- It is **scalable**, **flexible**, and **efficient**, but it is *complex and difficult to develop and maintain*. It is *used in medium to large systems* that *require high performance and reliability*. Adv. Prog.



Peer-to-Peer System Architecture

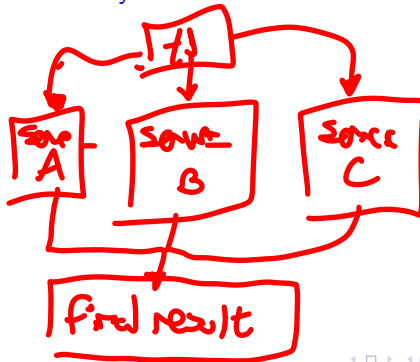
- A **peer-to-peer system architecture** is a **two-tier architecture** that consists of a **network of peers** that *communicate* with each other directly.
- It is **scalable**, **flexible**, and **efficient**, but it is **complex** and *difficult* to develop and maintain. It is *used* in **medium to large systems** that require **high performance** and **reliability**.

- Global Networks
- Domestic
- Computing Cluster
- Kubernetes



Distributed System Architecture

- A **distributed system architecture** is a **multi-tier architecture** that consists of a **network of nodes** that *communicate* with each other over a **network**.
- It is **scalable**, **flexible**, and **efficient**, but it is **complex** and **difficult** to develop and maintain. It is *used in* **large systems** that *require* high performance and **reliability**.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

